

Beepul Bharti

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Research Interests

- Safe anytime-valid inference (e-processes, sequential testing, online learning)
- Uncertainty in learning: calibration, conformal prediction, etc.
- Theory of responsible AI: interpretability, explainability, and algorithmic fairness
- Applications: high-stakes decision areas such as governance and healthcare

Education

- 2020– Ph.D. in Biomedical Engineering, Johns Hopkins University
Advisor: Dr. Jeremias Sulam
Relevant Coursework: Statistics, Matrix Analysis, Causal Inference, Nonlinear Optimization, Sparsity in ML/CV, Learning Theory, Replicable ML, Probabilistic Models in Vision
- 2016–20 B.A. in Mathematics & B.S. in Biomedical Engineering, Duke University
Honors: Cum Laude, Departmental Distinction
Relevant Coursework: Real Analysis, Abstract Algebra, Ordinary & Partial Differential Equations, Fluid Dynamics, Biostatistics, Probability, Multivariable Calculus

Experience

- Sum' 2024 **Genentech**, Machine Learning Research Intern San Francisco, CA
Deep Learning Theory and Algorithms Lab (DELTA)
- Explainability for motif discovery in genomic neural networks
- Sum' 2017 **Duke University**, Bass Connections Fellow Durham, NC
Department of Biostatistics
- Machine learning to predict schizophrenia admittance

Publications (*Indicates Equal Contribution)

- 2026 Parameter-Free and Group Conditional Online Conformal Prediction
Beepul Bharti, Ambar Pal, Jacopo Teneggi, Jeremias Sulam
Under review.
- 2026 Global Sequential Testing for Multi-Stream Auditing
Beepul Bharti, Ambar Pal, Jeremias Sulam
Under review.
- 2026 Let Me Explain, Again: Multiplicity in Local Sufficient Explanations
Ryan Pilgrim*, Beepul Bharti*, Jeremias Sulam, Rene Vidal
Transactions on Machine Learning Research (TMLR).
- 2025 Multiaccuracy and Multicalibration via Proxy Groups
Beepul Bharti, Mary Versa Clemens-Sewall, Paul Yi, Jeremias Sulam.
ICML.
- 2025 Uncovering BioLOGICAL Motifs and Syntax via Sufficiency and Necessary Explanations
Beepul Bharti, Gabrielle Scalia, Alex Tseng.
ICLR Workshop on Machine Learning for Genomics Explorations.

- 2024 Sufficient and Necessary Explanations (and What Lies in Between)
Beepul Bharti, Paul Yi, Jeremias Sulam.
Conference on Parsimony and Learning (CPAL). **Oral**.
- 2024 Evaluation of AI Algorithmic Biases in Radiology
Paul H Yi, Preetham Bachina, Beepul Bharti, Sean P Garin, Adway Kanhere, Pranav Kulkarni, David Li, Vishwa S Parekh, Samantha M Santomartino, Linda Moy, Jeremias Sulam.
Radiology.
- 2023 Estimating and Controlling for Equalized Odds via Sensitive Attributes
Beepul Bharti, Paul Yi, Jeremias Sulam.
NeurIPS.
- 2023 SHAP-XRT: The Shapley Value Meets Conditional Independence Testing
Jacopo Teneggi*, Beepul Bharti*, Yaniv Romano, Jeremias Sulam.
Transactions on Machine Learning Research (TMLR).
- 2023 Sex imbalance produces biased deep learning models for knee osteoarthritis detection
David Li, Beepul Bharti, Jinchi Wei, Jeremias Sulam, Paul H Yi.
Canadian Association of Radiologists Association.

Honors & Awards

- 2023 Alpha Eta Mu Beta: National Biomedical Engineering Honor Society
- 2022 JHU Mathematical Institute for Data Science Fellowship
- 2020 JHU Institute of Computational Medicine Fellowship
- 2016–2020 Duke University Dean's List
- 2019–2020 Duke University Pratt Fellowship
- 2019 Tau Beta Pi: The Engineering Honor Society

Teaching

Teaching Assistant

- 2023 EN.580.69: Biomedical Data Design, JHU
- 2021 EN.580.697: Computational Cardiology, JHU
- 2018 ECE 110L: Fundamentals of Electrical and Computer Engineering, Duke

Presentations

Talks

- 2026 Global Sequential Testing for Multi-Stream Auditing
UPenn CS Theory Seminar
- 2025 Sufficient and Necessary Explanations (and What Lies in Between)
Second Conference on Parsimony and Learning
- Sufficient vs. Necessary Explanations
Machine Learning in Healthcare Club, University of North South Wales
- 2024 SHAP-XRT: The Shapley Value Meets Conditional Independence Testing
Explainable AI Seminars at Imperial College London
- Algorithmic Fairness in Machine Learning and Data Science
EN.540.464: Advanced Biomedical Data Science for Biomedical Engineering

- 2023 Evaluating Fairness of AI Models in Radiology
Radiological Society of North America (RSNA) Annual Meeting
- Fairness in Machine Learning
EN.540.464: Advanced Biomedical Data Science for Biomedical Engineering
- 2022 Shapley Values and Hypothesis Testing
QMUL Intelligent Sensing Winter School
- Posters**
- 2026 Global Sequential Testing for Multi-stream Auditing
Safe Anytime-Valid Inference (SAVI) Conference
- 2024 Certifying Fairness with Incomplete Sensitive Information
SIAM Conference on Mathematics of Data Science
- 2023 Fairness via Sensitive Attribute Predictors
Columbia University Workshop on Fairness in Operations and AI
- Fairness with Missing Sensitive Attributes
Johns Hopkins AI-X Foundry Fall Symposium
- Shapley Values and Hypothesis Testing
Bern Interpretable AI Symposium
- 2018 Using Machine Learning to Predict Schizophrenia Admittance
Duke School of Medicine Clinical Research Day

Service

- 2025 Foreign Exchange Student Mentor
JHU Foreign Exchange Student Program
- 2024 High School Student Mentor
JHU Whiting School of Internships in Science and Engineering (WISE)
- 2023 Mentor Liaison
JHU Biomedical Engineering Application Assistance Program (BMEAAP)

Reviewing

Reviewer for ICLR, ICML, TMLR, FAccT, ISIT, AISTATS, NeurIPS, CPAL

Other

- Tools* Python, PyTorch, R
- Citizenship* United States
- Linguistics* English, Hindi (fluent)
- Highlights* 2017 TEDx Duke Speaker